DTU410, DTU420

GPRS/NB-IoT Data Loggers.
MQTT Getting Started Guide

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Contents

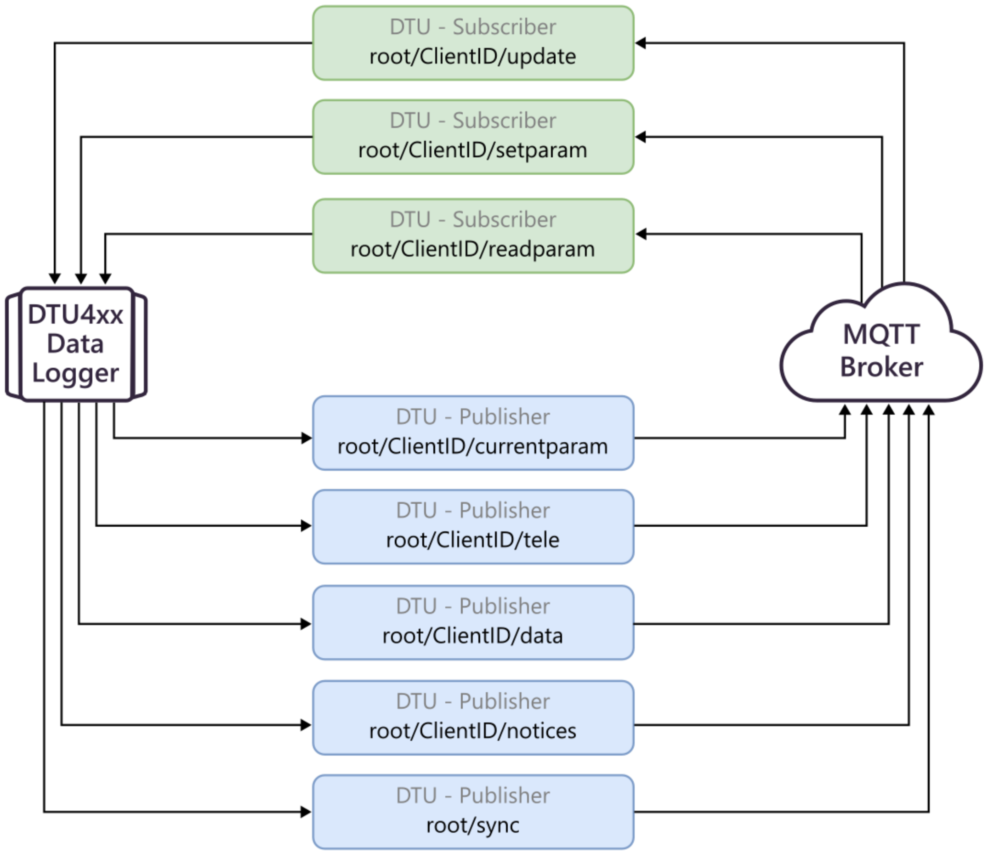
Overview

MQTT is an open ISO standard providing a lightweight messaging protocol for small sensors and mobile devices and optimized for high-latency or unreliable networks. The protocol is particularly widely used for IoT/M2M communications.

DTU4xx data loggers provide support for MQTT v5.0 protocol:

- **DTU410 GPRS data loggers** – starting from **DTU42.00.0007** firmware version
- **DTU420 NB-IoT data logger** – starting from **DTU42.00.0015** firmware version

In this guide, you will find necessary information on using the MQTT protocol with DTU devices.



Initial Data logger Configuration

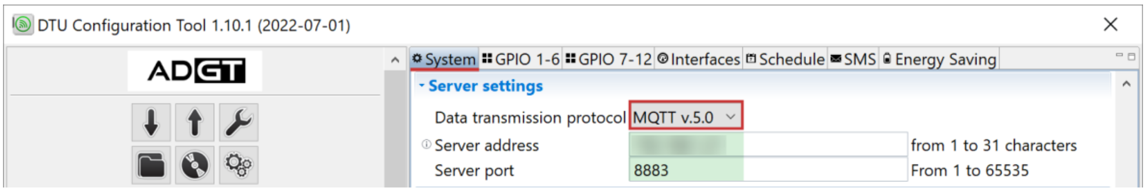
PLEASE NOTE: Before you start configuring DTU data loggers to enable MQTT, you need to have a working MQTT broker. You may use any convenient broker, for example, an open-source Mosquitto.

When setting up data logger, you can choose which transmission protocol to use: **ADGT DTU** (proprietary, set by default) or **MQTT v5.0**. You can configure the protocol settings via a PC using **DTU Configuration Tool** software, starting from version **1.10.1**.

To enable MQTT protocol for data logger:

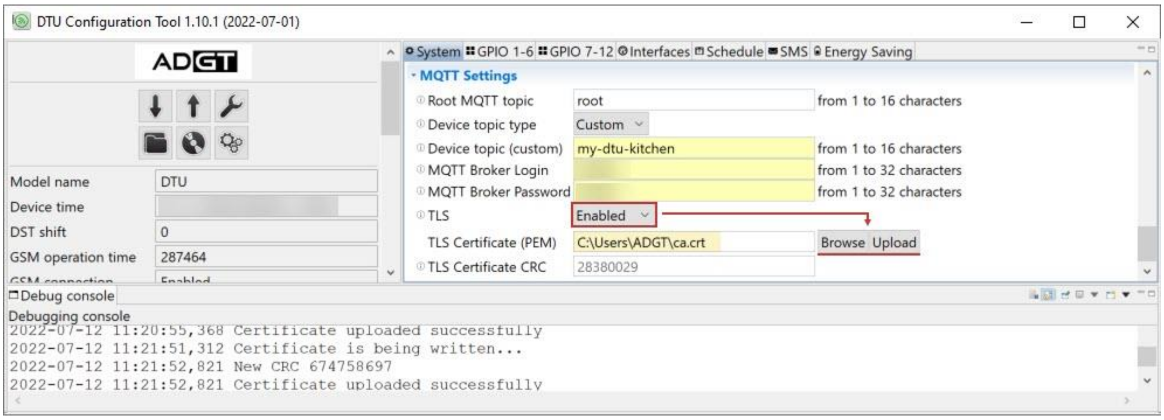
1. Connect device to PC via USB port and run **DTU Configuration Tool** program.

2. On the **System** tab -> **Server Settings**, select data transmission protocol – **MQTT v.5.0**. Set up the server address and port.



3. On the **System** tab -> **MQTT Settings**, set up MQTT Client settings for data logger:

MQTT Client Parameter	Description	Default value
Root MQTT topic	Root topic name. User-configurable, case sensitive	**root**
Device topic type	A placeholder for MQTT ClientID: – DTU IMEI number (set by default); – custom (user-configurable, case sensitive). <u>Example</u>: If you set device topic as ***my-dtu-kitchen***, the topic path will be ***root/my-dtu-kitchen*** <u>Please note</u>: If a device with such topic type already exists on the server, the connection error "*Invalid ClientID*" will occur.	IMEI number MQTT path: **root/[ClientID]**
MQTT Broker Login	Login for authentication on MQTT server.	not set
MQTT Broker Password	Password for authentication on MQTT server.	not set
TLS	Enables secure SSL/TLS connection (recommended).	disabled
TLS Certificate (PEM)	Upload your TLS certificate to this field.	not set
TLS Certificate CRC	A checksum, that verifies the authenticity of the certificate. Calculated automatically when the certificate is loaded (*not configurable*).	not set



4. Click **Write settings** button to save the settings, after which the device will restart automatically.

   All unsaved settings will be highlighted in yellow.

MQTT Message Format

Message format (from DTU to Broker and from Broker to DTU): **JSON**

Events start with the text header "event", for example: "***event***" : "**restart**", i.e. the event is a restart.

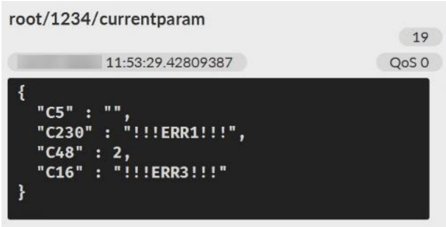
Parameters start with "***Cabc***" (from 1 to 3 digits), where
"*abc*" is the parameter number specified in **ADGT DTU Data transmission protocol**.
Example: **C24** is the 24th parameter (*Resistance of the normally closed contact_Input 3*).

The local time is transmitted in the following format:
"**time**" : "**YYYY-MM-DDThh:mm:ss**"
Example: ***time : "2021-06-20T18:45:21"***, i.e. June 20, 2021, 18:45:21

In addition to the local time, the timezone label is reported in the following format: "**tz**": "**±hh:mm**"

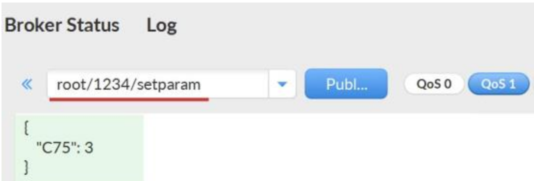
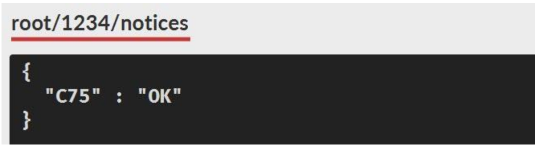
Codes for executing commands when configuring the device:

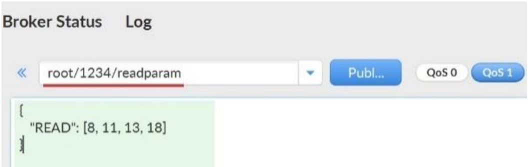
Code	Description
OK	Command is executed
ERR1	Command is not supported
ERR2	Invalid data format
ERR3	Other errors
Error code format: "!!!***ERR_CODE***!!!" Example: "C20" : "**!!!ERR1!!!**"	

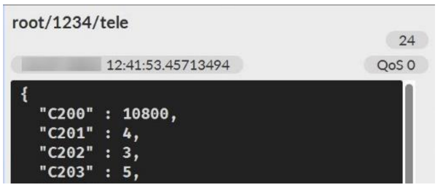
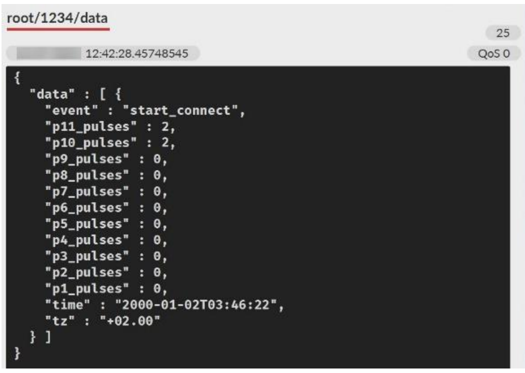


MQTT Topics

PLEASE NOTE: Level of service for all DTU topics – **QoS 1** (at least once), guarantees that a message is delivered at least one time to the receiver.

Description	MQTT Topic	DTU Function
1. Sending settings from MQTT broker to DTU data logger		
User can remotely change the DTU settings by publishing a JSON file with the settings to the topic **root/[ClientID]/setparam** via MQTT broker software. 	root/[ClientID]/setparam	Subscriber
Data logger publishes message on the application of settings to the topic **root/[ClientID]/notices** 	root/[ClientID]/notices	Publisher

<table>
<tr><th>Description</th><th>MQTT Topic</th><th>DTU Function</th></tr>
<tr><td colspan="3">2. Transmitting DTU current settings to MQTT broker</td></tr>
<tr><td>User publishes a JSON request via the broker's software,
to read DTU settings in the topic root/[ClientID]/readparam
Request format:
{"READ": [8, 11, 13, 18]}, where:
– READ - general command to read the settings
– [8, 11, 13, 18] - enumeration of required parameter numbers, without "C"

</td><td>root/[ClientID]/readparam</td><td>Subscriber</td></tr>
<tr><td>At each connection, data logger checks the topic root/[ClientID]/readparam,
if there is a request there, it publishes a file with the requested settings in root/[ClientID]/currentparam

</td><td>root/[ClientID]/currentparam</td><td>Publisher</td></tr>
<tr><td colspan="3">3. Sending telemetry data to MQTT broker for analytics</td></tr>
<tr><td>Data logger, while connecting to the server on a schedule, publishes a json telemetry file in the topic root/[Client ID]/tele
NOTE: If it is necessary to send a large number of parameters, files are divided into parcels of <u>20 parameters</u> (not more)

</td><td>root/[Client ID]/tele</td><td>Publisher</td></tr>
<tr><td colspan="3">4. Transmission of measured values & events to MQTT broker during regular connection</td></tr>
<tr><td>Data logger, while connecting to the server on a schedule, publishes raw input values and events in root/[Client ID]/data

</td><td>root/[Client ID]/data</td><td>Publisher</td></tr>
</table>

Description	MQTT Topic	DTU Function
5. Data transmission in case of emergency connection		
In case of emergency situations, DTU connects to the server outside the schedule and publishes data to the relevant topics. The list of transmitted data/parameters is similar to those that are transmitted during emergency connection when working in the **ADGT DTU Protocol**	Topics to which messages can be sent: root/[Client ID]/tele root/[Client ID]/data	Publisher
6. Device time synchronization with MQTT broker		
By default, data logger is subscribed to the topic for time synchronization with the MQTT server – **root/sync**. We offer a synchronization solution - a script that is run on the MQTT broker's PC and publishes the current UTC time in **root/sync** every 5 seconds with QoS=0. Time recording format: **{"UTCTime" : "YYYY-MM-DDThh:mm:ss"}** When connecting to the server, the DTU data logger begins to receive timestamps until the data transfer session ends. • If the data logger receives 2 consecutive timestamps with an interval of 5 to 7 seconds, then the time is synchronized according to the last timestamp. • If the interval between the two received timestamps is more than 7 seconds, then the packets arrived with a delay, synchronization does not occur. Data logger is waiting for the next timestamps at the required interval. The .zip file with the script can be downloaded from our site https://adgt.cz **NOTE:** You can use your own solution to sync the time.	root/sync	Publisher

MQTT Event Codes

ADGT DTU Code	Description	MQTT code
1	Data slicing event (time interval has passed)	time_passed
2	ADC event (break or short circuit)	adc_break, adc_short
3	Device restart	restart
4	Dry contact triggering	dry_contact_on
8	S2 button was pressed	start_connect
12	GPRS (NB-IoT) connection failed	gprs_connect_failed
13	External power has been disabled	ext_pwr_lost
14	External power has been re-enabled	ext_pwr_on
15	Pulse input polling frequency is exceeded	pulses_exceed
16	End of the archive transmission (not saved in the journal)	archv_transfrd
17	Maximum SIM card inactivity period is exceeded	connect_timeout
19	Input value monitoring: **event**	input_val_ctrl
20	Permissible excess of the values at the inputs is exceeded	input_val_exceed

ADGT DTU Code	Description	MQTT code
21	Pulse input polling frequency restored	pulses_restored
22	Battery is depassivated	bat_depassivated
23	Battery is low	bat_discharged
24	Current loop input value is too high	current_loop_toohigh
25	Current loop input value is too low	current_loop_toolow
26	Current loop input values are restored	current_loop_restored

MQTT Data Types

<table>
<tr><th>Number</th><th>Size, byte</th><th colspan="2">Description</th><th>MQTT Code</th></tr>
<tr><td>0</td><td>4</td><td colspan="2">Meter 1 value, in pulses (Data interpretation)</td><td>p1_pulses</td></tr>
<tr><td>1</td><td>4</td><td colspan="2">Meter 2 value, in pulses (Data interpretation)</td><td>p2_pulses</td></tr>
<tr><td>2</td><td>4</td><td colspan="2">Meter 3 value, in pulses (Data interpretation)</td><td>p3_pulses</td></tr>
<tr><td>3</td><td>4</td><td colspan="2">Meter 4 value, in pulses (Data interpretation)</td><td>p4_pulses</td></tr>
<tr><td>6</td><td>4</td><td colspan="2">Number of data logger restarts</td><td>restarts_count</td></tr>
<tr><td>7</td><td>1</td><td>Input 1 state</td><td rowspan="4">low – Logical "0"
short – Short Circuit
break – Break
high – Logical "1"</td><td>p1_state_low
p1_state_short
p1_state_break
p1_state_high</td></tr>
<tr><td>8</td><td>1</td><td>Input 2 state</td><td>p2_state_low
p2_state_short
p2_state_break
p2_state_high</td></tr>
<tr><td>9</td><td>1</td><td>Input 3 state</td><td>p3_state_low
p3_state_short
p3_state_break
p3_state_high</td></tr>
<tr><td>10</td><td>1</td><td>Input 4 state</td><td>p4_state_low
p4_state_short
p4_state_break
p4_state_high</td></tr>
<tr><td>12</td><td>4</td><td colspan="2">Input 1 resistance in closed state</td><td>p1_short_res</td></tr>
<tr><td>13</td><td>4</td><td colspan="2">Input 1 resistance in open state</td><td>p1_open_res</td></tr>
<tr><td>14</td><td>4</td><td colspan="2">Input 2 resistance in closed state</td><td>p2_short_res</td></tr>
<tr><td>15</td><td>4</td><td colspan="2">Input 2 resistance in open state</td><td>p2_open_res</td></tr>
<tr><td>16</td><td>4</td><td colspan="2">Input 3 resistance in closed state</td><td>p3_short_res</td></tr>
<tr><td>17</td><td>4</td><td colspan="2">Input 3 resistance in open state</td><td>p3_open_res</td></tr>
<tr><td>18</td><td>4</td><td colspan="2">Input 4 resistance in closed state</td><td>p4_short_res</td></tr>
<tr><td>19</td><td>4</td><td colspan="2">Input 4 resistance in open state</td><td>p4_open_res</td></tr>
<tr><td>20</td><td>1</td><td colspan="2">Server Connection Error Codes</td><td>connect_error</td></tr>
<tr><td>21</td><td>4</td><td colspan="2">Processor power voltage (mV)</td><td>cpu_voltage</td></tr>
<tr><td>22</td><td>1</td><td colspan="2">Input number (#) where polling frequency is exceeded</td><td>port#consumption_exceeded</td></tr>
<tr><td>23</td><td>1</td><td colspan="2">SIM card number (#), where GSM error occurred
(0 - SIM1, 1 - SIM2)</td><td>error_sim#</td></tr>
</table>

Number	Size, byte	Description	MQTT Code
24	1	SIM (#) with inactivity period exceeded (0 - SIM1, 1 - SIM2)	inactive_sim#
25	1	Input 5 state low – Logical “0” short – Short Circuit break – Break high – Logical “1”	p5_state_low p5_state_short p5_state_break p5_state_high
26	1	Input 6 state	p5_state_low p5_state_short p5_state_break p5_state_high
27	4	Input 5 resistance in closed state	p5_short_res
28	4	Input 5 resistance in open state	p5_open_res
29	4	Input 6 resistance in closed state	p6_short_res
30	4	Input 6 resistance in open state	p6_open_res
31	1	Input number (#) where value is above/below the threshold	port#out_of_threshold
32	1	Input event was triggered: 0 - below the specified limit, 1 - inside the specified limit, 2 - above the specified limit	delta_event
33	1	Input number (#) where the value exceeded the base value by a delta	port#exceeded_delta
37	4	Input IN1: meter value, in pulses (see Data interpretation)	p5_pulses
38	4	Input IN2: meter value, in pulses	p6_pulses
39	4	Input IN3: meter value, in pulses	p7_pulses
40	4	Input IN4: meter value, in pulses	p8_pulses
41	4	Input IN5: meter value, in pulses	p9_pulses
42	4	Input IN6: meter value, in pulses	p10_pulses
43	4	Meter value for input S (leakage), in pulses	p11_pulses
44	1	Input IN1/P5 state: low - Logical 0, short - Short Circuit, break - Break, high - Logical 1	p5_state_low p5_state_short p5_state_break p5_state_high
45	1	Input IN2/P6 state: low - Logical 0, short - Short Circuit, break - Break, high - Logical 1	p6_state_low p6_state_short p6_state_break p6_state_high
46	1	Input IN3/P7 state: low - Logical 0, short - Short Circuit, break - Break, high - Logical 1	p7_state_low p7_state_short p7_state_break p7_state_high
47	1	Input IN4/P8 state: low - Logical 0, short - Short Circuit, break - Break, high - Logical 1	p8_state_low p8_state_short p8_state_break p8_state_high
48	1	Input IN5/P9 state: low - Logical 0, short - Short Circuit, break - Break, high - Logical 1	p9_state_low p9_state_short p9_state_break p9_state_high
49	1	Input IN6/P10 state: low - Logical 0, short - Short Circuit, break - Break, high - Logical 1	p10_state_low p10_state_short p10_state_break p10_state_high
50	4	The value of the battery voltage at a reference load (mV)	bat_voltage

Data Interpretation in MQTT

Input Type	Data Description	MQTT code
Pulse Counter	Four-byte unsigned value, displaying the number of pulses passed through the current input.	pulses
Temperature Sensor	Divided into four signed bytes (from -128 to 128 bytes). Shows the temperature value measured by the external analog sensor: • First byte – current temperature • Second byte – average temperature for the latest data slice • Third byte – minimum temperature for the latest data slice • Fourth byte – maximum temperature for the latest data slice	temp
DS18B20 (1-Wire) Sensor	Divided into two signed two-byte values (from -32767 to 32767 bytes). Shows the temperature value (*in tenths of a degree Celsius*), measured by the digital DS18B20 temperature sensor. • First two bytes – current temperature • Second two byte – average temperature for the latest data slice **Attention! Temperature measurements are taken every five minutes**	tempds
Engine Hours Counter	Four-byte unsigned value, indicates how long the input was in the active state (*in sec*).	bat_time
4-20mA Current Loop Sensor	Four-byte unsigned value, indicates the current value in microamperes (μA).	current

Server Connection Error Codes

Error #	Description	MQTT Code
0	The session passed without errors	no_error
1	Incorrect PIN	need_pincode
2	SIM card is not inserted	no_sim
3	GSM network registration failed	no_registration
4	GPRS (NB-IoT) connection failed	no_gprs_connect
5	Connection to server failed	no_server_connect

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